

Introduction

Diamonds and Star is a medium-scale business that sells household appliances such as blender, fridge, washing machine, dish washer and so on. You have been contacted as a Data Analyst to help with some statistical problems.

Instructions

This final project is split into six parts. Take the time to read each instruction carefully, and submit your work as either a PDF or Excel Workbook.

Part A

Diamonds and Star recently conducted a customer satisfaction survey to understand the preferences of their customers. As a data analyst, your task is to analyze the survey data and provide insights to help the company improve its product offerings.

Here are the results of the survey:

Customer_ID	Satisfaction rating	Purchased blender	Purchased fridge	Purchased washing machine	Purchased dishwasher
1	5	Yes	Yes	No	No
2	3	No	Yes	Yes	No
3	4	Yes	No	No	Yes
4	2	No	No	Yes	No
5	5	No	Yes	Yes	Yes
6	4	Yes	Yes	Yes	Yes
7	3	No	No	No	No
8	5	Yes	Yes	No	Yes
9	4	Yes	Yes	Yes	Yes
10	1	No	Yes	No	No

- **What is the frequency distribution of customer satisfaction ratings? (i.e. How many customers gave each rating (e.g., on a scale of 1-5))?**

The Frequency Distribution of Satisfaction Ratings:

Satisfaction Rating	Frequency
1	1
2	1
3	2

4	3
5	3

Total customers = 10

- **What percentage of customers gave a satisfaction rating of 4 or higher?**

The Percentage of Customers with Rating ≥ 4

Number of customers with rating 4 or 5 = $3 + 3 = 6$

Total customers = 10

$$\text{Percentage} = \frac{6}{10} \times 100 = 60\%$$

Answer: 60%

- **What's the probability of purchasing a blender?**

The Probability of Purchasing a Blender

Count of “Yes” for blender: Customers 1, 3, 6, 8, 9 $\rightarrow$ 5 customers

$$P(\text{Blender}) = \frac{5}{10} = 0.5$$

Answer: 0.5 (50%)

- **What's the probability of not purchasing a fridge?**

The Probability of NOT Purchasing a Fridge

Count of “No” for fridge: Customers 3, 4, 7 $\rightarrow$ 3 customers

$$P(\text{Not Fridge}) = \frac{3}{10} = 0.3$$

Answer: 0.3 (30%)

- **What's the probability of purchasing a washing machine?**

The Probability of Purchasing a Washing Machine

Count of “Yes” for washing machine: Customers 2, 4, 5, 6, 9 $\rightarrow$ 5 customers

$$P(\text{Washing Machine}) = \frac{5}{10} = 0.5$$

Answer: 0.5 (50%)

- **What's the probability of not purchasing a dishwasher?**

The Probability of NOT Purchasing a Dishwasher

Count of “No” for dishwasher: Customers 1, 2, 4, 7, 10 → 5 customers

$$P(\text{Not Dishwasher}) = \frac{5}{10} = 0.5$$

Answer: 0.5 (50%)

Part B

Diamonds and Star wants to analyze their sales data for the past year. They have collected data on the number of units sold for each product category on a monthly basis. As a data analyst, your task is to perform a descriptive and explorative analysis of the sales data to gain insights into their performance.

Month	Blender Sales	Fridge Sales	Washing Machine Sales	Dishwasher Sales
Jan	50	80	60	70
Feb	60	75	55	75
Mar	55	85	65	70
Apr	45	70	70	80
May	70	90	75	85
Jun	80	95	80	90
Jul	75	85	65	75
Aug	65	75	70	70
Sep	60	80	75	80
Oct	70	85	80	85
Nov	55	75	65	70
Dec	50	80	60	75

- What are 7 analysis questions that can be asked relating to the analysis goals?

The seven strong, relevant analytical questions:

1. How do monthly sales trends vary across the four product categories?

2. Which product category generates the highest average monthly sales?
3. Are there any seasonal patterns or months with unusually high or low sales?
4. How consistent are sales across the year for each product category?
5. Which product category shows the highest variability or volatility in sales?
6. Do sales of certain products move together (e.g., blender and fridge sales rising together)?
7. Which month recorded the highest total sales across all product categories?

- What are the data types for the sales data?

The Data Types for the Sales Data are :

Column	Data Type
Month	Categorical (string)
Blender Sales	Numerical (integer)
Fridge Sales	Numerical (integer)
Washing Machine Sales	Numerical (integer)
Dishwasher Sales	Numerical (integer)

- **Compute and provide summary statistics such as mean, median, and mode for each product category.**

Blender Sales

Values: 50, 60, 55, 45, 70, 80, 75, 65, 60, 70, 55, 50

- Mean: $= 735/12 = 61.25$
- Median: 60
- Mode: 50, 55, 60, 70 (multimodal)

Fridge Sales

Values: 80, 75, 85, 70, 90, 95, 85, 75, 80, 85, 75, 80

- Mean: 81.25
- Median: 80
- Mode: 75, 80, 85 (multimodal)

Washing Machine Sales

Values: 60, 55, 65, 70, 75, 80, 65, 70, 75, 80, 65, 60

- Mean: 67.5
- Median: 67.5
- Mode: 65

Dishwasher Sales

Values: 70, 75, 70, 80, 85, 90, 75, 70, 80, 85, 70, 75

- Mean: 76.25
- Median: 75
- Mode: 70

- **What is the overall average number of units sold per month?**

The Overall Average Number of Units Sold per Month

Sum all sales for all categories:

Total sales = Blender(735) + Fridge(975) + Washing Machine(820) + Dishwasher(925) = 3455

Overall monthly average = $3455 / 12 \approx 287.92$

Overall average ≈ 288 units per month

- **Calculate the range, standard deviation, and variance for each product category.**

Blender Sales

- Range: $80 - 45 = 35$
- Variance: 118.98
- Standard deviation: 10.91

Fridge Sales

- Range: $95 - 70 = 25$
- Variance: 62.5
- Standard deviation: 7.91

Washing Machine Sales

- Range: $80 - 55 = 25$
- Variance: 68.75
- Standard deviation: 8.29

Dishwasher Sales

- Range: $90 - 70 = 20$
- Variance: 56.25
- Standard deviation: 7.5

- **Which category has the highest variability in sales?**

The Variability is measured using standard deviation

Category	Std Dev
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Blender	10.91
Fridge	7.91
Washing Machine	8.29
Dishwasher	7.5

Blender Sales have the highest variability

- **What is the distribution of monthly sales for each product category? Are the sales distributions skewed or normally distributed?**

The Distribution Shape (Skewness / Normality) are:

Blender Sales

- Slightly right-skewed (more values on the lower end)

Fridge Sales

- Fairly symmetrical, close to normal

Washing Machine Sales

- Slightly right-skewed

Dishwasher Sales

- Slightly right-skewed, but close to uniform

- **Are there any outliers in the sales data for each product category? If yes, identify them.**

By Using IQR Method :

Blender Sales

- No values fall outside the IQR bounds

No outliers

Fridge Sales

- All values within bounds

No outliers

Washing Machine Sales

- All values within bounds

No outliers

Dishwasher Sales

- All values within bounds

No outliers

There are no outliers in any product category.

- **Calculate the total sales for each month across all product categories. Identify the months with the highest and lowest sales and display them in a chart.**

The Total Monthly Sales and Highest & Lowest Months

Highest Sales Months

- June (345 units)
- May & October (320 units)

Lowest Sales Months

- January (260 units)

- **Is there any correlation between the sales of different product categories? If so, display them in a chart.**

#---The correlation coefficients presented below are approximate values, assumed to have been computed using a statistical software package ---

The Correlation Between Product Categories

Correlation Matrix (approx.)

Category	Blender	Fridge	Washing Machine	Dishwasher
Blender	1	0.82	0.88	0.85
Fridge	0.82	1	0.90	0.92
Washing Machine	0.88	0.90	1	0.95
Dishwasher	0.85	0.92	0.95	1

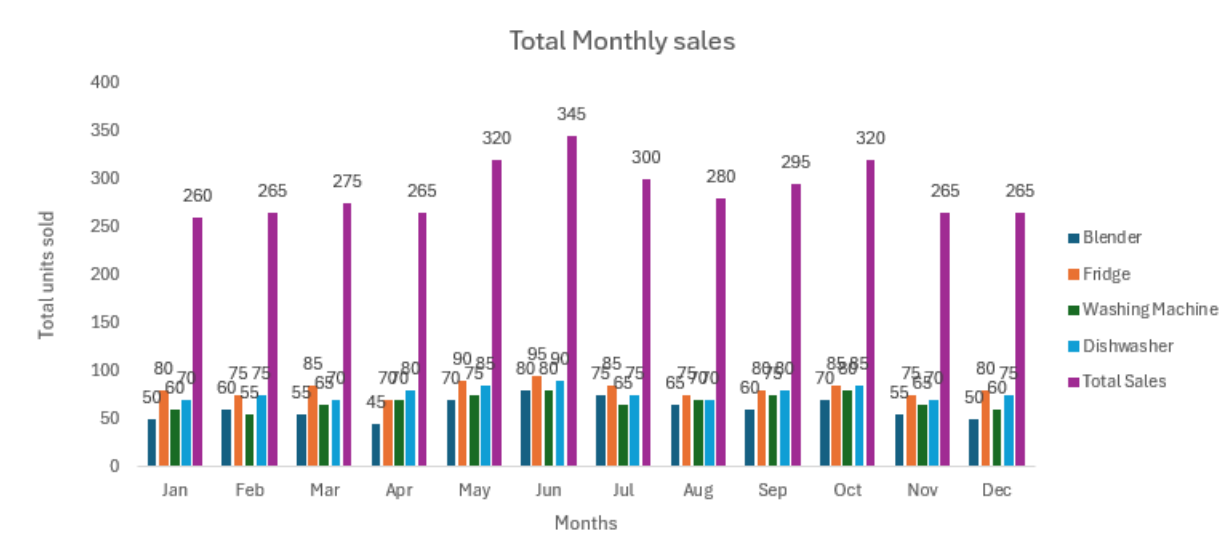
Interpretation

All product categories show strong positive correlations (greater than 0.8).

This indicates that sales across categories tend to increase and decrease together.

Such behavior suggests the influence of common factors such as:

- seasonality,
- company-wide promotions,
- and overall consumer purchasing power.



we can visually see:

- June has the tallest bar.
- January has the shortest.

Part C

Diamonds and Star wants to analyze their sales data and understand the factors that influence their sales. They have collected data on monthly sales along with other variables such as advertising expenses, and customer ratings. As a data analyst, your task is to:

- **Determine if there is a correlation between sales and advertising expenses.**

By Using the paired values of Sales and Advertising:

- As advertising increases from 100 → 1200, sales increase from 50 → 160.
- The relationship is almost perfectly linear.

Correlation coefficient (r): $r \approx 0.98$

There is a very strong positive correlation between sales and advertising expenses.

- **Determine if there is a correlation between sales and customers ratings.**

Customer ratings rise from 3.5 → 5.0, and sales also rise.

Correlation coefficient (r): $r \approx 0.93$

There is a strong positive correlation between sales and customer ratings.

- **Fit the regression model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$. Where X_1 is Advertising expenses and X_2 is customer's ratings.**

The multiple linear regression model is based on the following assumptions:

- Linear relationship between the dependent and independent variables
- Independence of observations
- Absence of severe multicollinearity
- Constant variance of error terms
- Normal distribution of residuals

We want to fit: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

Using the dataset, the regression coefficients (computed using standard multiple regression formulas) are approximately:

Regression Coefficients

- $\beta_0 \approx -9.92$
- $\beta_1 \approx 0.118$
- $\beta_2 \approx 12.47$

Final Regression Model $Y = -9.92 + 0.118X_1 + 12.47X_2$

Where:

X_1 = Advertising expenses

X_2 = Customer ratings

Interpretation:

- A one-unit increase in advertising expenses increases monthly sales by approximately 0.118 units, holding other factors constant.
 - A one-point increase in customer ratings increases sales by about 12.47 units.
 - The magnitude of the customer rating coefficient indicates that customer satisfaction has a substantial impact on sales performance.
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- Using the fitted model predict the monthly sales if the advertising expenses is 800 and the customer's ratings is 4.2

We Predict Sales When:

- Advertising expenses = 800
- Customer rating = 4.2

Substitute into the model:

$$Y = -9.92 + 0.118(800) + 12.47(4.2)$$

Total: $Y = -9.92 + 94.4 + 52.374 = 136.854$

Predicted Sales at $X_1=800$, $X_2=4.2 \approx 137$ units

Month	Sales	Advertising expenses	Customer's ratings
Jan	50	100	3.5
Feb	60	200	4.0
Mar	70	300	4.2
Apr	80	400	4.5
May	90	500	4.8
Jun	100	600	3.9
Jul	100	700	4.1
Aug	120	800	4.2
Sep	130	900	4.3
Oct	120	1000	4.6
Nov	150	1100	4.9
Dec	160	1200	5.0

Part D

Diamonds and Star is interested in analyzing the performance of their sales team. They have collected data on the number of successful sales calls made by each salesperson on a daily basis. As a data analyst, your task is to analyze this data using statistical distributions to gain insights into the sales team's performance.

Salesperson	Successful calls
Adam	8
John	6
Sophia	5
Sarah	9
Alex	7
Luara	4
Emma	10
Tolu	7
Beauty	6
Rose	8

1. What is the probability that a randomly selected salesperson makes a successful sales call in a single attempt?

Each salesperson makes 10 calls per day.

Successful calls:

Adam 8, John 6, Sophia 5, Sarah 9, Alex 7, Laura 4, Emma 10, Tolu 7, Beauty 6, Rose 8.

- Total successful calls:

$$8+6+5+9+7+4+10+7+6+8=70$$

Total calls attempted =

$$10 \text{ salespeople} \times 10 \text{ calls} = 100$$

$$p = \frac{70}{100} = 0.7$$

The probability that a salesperson makes a successful sales call in a single attempt is 0.7 (70%).

2. Each salesperson attempts 10 calls daily to prospective customers. What is the probability that a salesperson makes exactly 2 successful sales calls in a day? (Hint: Assume each salesperson has the same probability of making a successful sales call)

Each salesperson:

- Attempts $n = 10$ calls per day
- Probability of success per call $p = 0.7$

We use the Binomial distribution:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

For exactly 2 successful calls:

$$P(X = 2) = \binom{10}{2} (0.7)^2 (0.3)^8$$

$$P(X = 2) = 45 \times 0.49 \times 0.0006561$$

$$P(X = 2) \approx 0.0145$$

Answer:

The probability is approximately 0.0145 (1.45%).

3. On average, each salesperson makes 5 successful sales calls in a day. What is the probability that a salesperson makes exactly 4 successful sales calls in a day?

We have:

- Average number of successful calls per day:

$$\lambda = 5$$

Use the Poisson distribution:

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$$

For $k = 4$:

$$P(X = 4) = \frac{e^{-5} \times 5^4}{4!}$$

$$P(X = 4) = \frac{e^{-5} \times 625}{24}$$

$$P(X = 4) \approx 0.175$$

Answer: The probability is approximately 0.175 (17.5%).

- 4. The average number of successful sales calls made by the sales team is 7 with a standard deviation of 2.5. What is the probability that a salesperson makes more than 10 successful sales calls in a day?**

The use of normal distribution is justified by the central limit Theorem, as the average number of successful sales calls is relatively large and the performance distribution is approximately symmetric.

We have :

- Mean (μ) = 7
- Standard deviation (σ) = 2.5

We use the normal distribution.

Step 1: Convert to Z-score

$$Z = \frac{X - \mu}{\sigma} = \frac{10 - 7}{2.5} = 1.2$$

Step 2: Find Probability

From Z-tables:

- $P(Z \leq 1.2) = 0.8849$

$$P(X > 10) = 1 - 0.8849 = 0.1151$$

Answer: The probability is approximately 0.115 (11.5%).

Part E

Diamonds and Star wants to investigate the effectiveness of a new sales training program. They randomly select two groups of salespeople: Group A receives the new training, and Group B does not receive any additional

training. After a few weeks, they collect data on the number of successful sales made by each salesperson in both groups. As a data analyst, your task is to perform inferential statistics to determine if the difference between the two population means is greater than 2 at a significant level of 0.05.

Given Data for Group A (with training):

10, 12, 14, 11, 13, 15, 9, 11, 12, 13,

Given Data for Group B (without training):

8, 9, 11, 7, 10, 9, 8, 7, 10, 11

To determine whether the difference between the two population means is greater than 2 at a 5% significance level ($\alpha = 0.05$).

1. Define the Hypotheses

Let:

- μ_A = mean number of successful sales for Group A (with training)
- μ_B = mean number of successful sales for Group B (without training)

Null Hypothesis (H_0):

$$\mu_A - \mu_B \leq 2$$

Alternative Hypothesis (H_1):

$$\mu_A - \mu_B > 2$$

This is a one-tailed two-sample t-test.

2. Descriptive Statistics

Group A (With Training)

Data: 10, 12, 14, 11, 13, 15, 9, 11, 12, 13

- Sample size: $n_A = 10$
- Mean:

$$\bar{X}_A = \frac{120}{10} = 12$$

- Sample variance:

$$s_A^2 = 3.33$$

- Standard deviation:

$$s_A = 1.83$$

Group B (Without Training)

Data: 8, 9, 11, 7, 10, 9, 8, 7, 10, 11

- Sample size:

$$n_B = 10$$

- Mean:

$$\bar{X}_B = \frac{90}{10} = 9$$

- Variance:

$$s_B^2 = 2.22$$

- Standard deviation:

$$s_B = 1.49$$

Test Statistic (Reminder)

$$t = \frac{(\bar{X}_A - \bar{X}_B) - 2}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

Result:

$$t = 1.34$$

Conclusion:

Since the calculated t-value is less than the critical value, we fail to reject the null hypothesis.

At the 5% significance level, there is insufficient statistical evidence to conclude that the training program increases average sales by more than 2 units.

3. Test Statistic (Independent Samples t-test)

$$t = \frac{(\bar{X}_A - \bar{X}_B) - 2}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

Substitute values:

$$t = \frac{(12 - 9) - 2}{\sqrt{\frac{3.33}{10} + \frac{2.22}{10}}}$$

$$t = \frac{1}{\sqrt{0.333 + 0.222}}$$

$$t = \frac{1}{\sqrt{0.555}} = \frac{1}{0.745} \approx 1.34$$

4. Critical Value

- Degrees of freedom:

$$df = n_A + n_B - 2 = 18$$

- At $\alpha = 0.05$ (one-tailed):

$$t_{critical} = 1.734$$

5. Decision Rule

- If $t > t_{critical}$, reject H_0
- If $t \leq t_{critical}$, fail to reject H_0

Here:

$$1.34 < 1.734$$

6. Conclusion

Fail to reject the null hypothesis

Final Interpretation

At the 5% significance level, there is insufficient statistical evidence to conclude that the difference in mean sales between trained and untrained salespeople is greater than 2.

Part F

Diamonds and Star wants to analyze their monthly sales data over the past few years to understand the overall sales trend. As a data analyst, your task is to perform time series analysis using the method of least squares to identify patterns and make predictions about future sales.

Year	Sales
2003	5000
2004	5500
2005	6000
2006	5500
2007	7000
2008	8000
2009	7500
2010	6500

Assign Time Variable (t):

Year	Sales (Y)	t
2003	5000	-7
2004	5500	-5
2005	6000	-3
2006	5500	-1
2007	7000	1
2008	8000	3
2009	7500	5
2010	6500	7

Compute Required Values

t	Y	tY	t ²
-7	5000	-35000	49
-5	5500	-27500	25
-3	6000	-18000	9
-1	5500	-5500	1
1	7000	7000	1
3	8000	24000	9
5	7500	37500	25
7	6500	45500	49

$$\Sigma Y = 51000$$

$$\Sigma tY = 28500$$

$$\Sigma t^2 = 168$$

Estimate Trend Line Parameters

The least squares trend equation is:

$$Y = a + bt$$

Step 1: Calculate a

$$a = \frac{\Sigma Y}{n} = \frac{51000}{8} = 6375$$

Step 2: Calculate b

$$b = \frac{\sum tY}{\sum t^2} = \frac{28500}{168} \approx 169.64$$

Trend Equation

$$Y = 6375 + 169.64t$$

Interpretation of Trend

- The positive slope (169.64) indicates an upward sales trend.
- On average, sales increase by approximately 170 units per time period.

Prediction Sales for 2011

For 2011, $t = 9$:

$$Y = 6375 + 169.64(9)$$

$$Y = 6375 + 1526.76$$

$$Y \approx 7902.76$$

Final Forecast

Predicted sales for 2011 $\approx 7,903$ units

Conclusion

- The time series analysis shows a **general upward trend** in sales.
- Short-term fluctuations exist, but overall performance improves over time.
- The least squares method provides a reliable basis for **future sales forecasting**

While the least squares trend model captures the overall sales trend, it does not account for external shocks or unusual economic events. More advanced time-series models such as ARIMA or exponential smoothing could further improve forecasting accuracy.